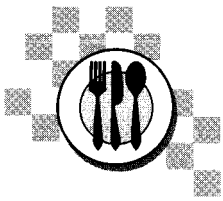
CHAPTER 17



Sanitary Food Handling in Foodservice

ABOUT THIS CHAPTER

In this chapter you will learn:

1. Why changes in the foodservice industry have made sanitation even more important
2. Three procedures that can help keep food from being contaminated
3. How clean preparation and serving areas, clean utensils, good employee health and hygiene, sanitary serving areas, and inspection programs can protect food
4. How to clean and sanitize surfaces and equipment in foodservice facilities
5. Why good management and employee training make a sanitation program work properly

INTRODUCTION

The United States has more than 710,000 foodservice establishments, employing about 10 million people. These foodservice operations range from mobile food stands, cafeterias, and fast-food chains to fancy restaurants. Consumers now spend almost half of their food budget on meals away from home. The foodservice industry has grown, and food production, processing, distribution, and preparation have changed. The major changes are more packaged foods, partly or fully prepared bulk foods, individual packages, and food production at a central location.

However much food production, handling, and preparation change, food can still be contaminated with microorganisms and can cause illness. Because of modern processing methods, handling, and distribution, it takes longer for food to reach the table, and it is more likely to be contaminated with microorganisms. Food served in restaurants is responsible for about 58% of outbreaks of foodborne illness. These

outbreaks cost somewhere between $1 billion and $10 billion. Centralized kitchens and mass feeding operations mean that more people are affected by a contaminated food. Protecting foodservice customers from foodborne illness is complicated but very important.

The main goal of a foodservice sanitation program is to protect the consumer and prevent or minimize food contamination. It is difficult to protect food from all contamination because pathogenic microorganisms are found almost everywhere and on about half the people who handle food.

SANITARY PROCEDURES FOR FOOD PREPARATION

Prepared food should be kept wholesome and safe by good sanitation during preparation and storage. Three sanitary procedures can help reduce contamination:

1. *Wash food.* Foodservice workers may not need to wash processed foods. However, they should rinse a can before opening it. They should wash all fresh fruits and vegetables before they are prepared (i.e., cut, sliced, peeled), eaten, or cooked, and also wash dried fruits and raisins, raw poultry, fish, and variety meats. Washing poultry reduces contamination of the inside of the poultry. Workers should wash fruits, vegetables, and meats with cold to lukewarm running water to remove dirt, and should drain all washed foods. Soaps and detergents are not used to wash foods because some residue will remain on the food after rinsing. If foods are not cooked immediately, they should be covered and refrigerated.

2. *Protect food.* It is important to protect food from contamination by bacteria that cause foodborne illness. Cleaning compounds, polishes, insect powders, and other compounds used in a foodservice operation can accidentally get into the food. Operators should store all chemicals away from food areas.

3. *Keep food hot or cold.* Processors should thoroughly cook and/or refrigerate all foods that can carry illness-producing microorganisms, including meat, poultry, fish, and dairy products. (In foodservice, most fruits and vegetables are also refrigerated. Cooked food, except some baked goods, is also refrigerated.) Various bacteria need different amounts of heat (different heating times and temperatures) to destroy them and their spores.

REDUCING CONTAMINATION

Preparation and Serving Areas

In food preparation and serving areas, air, surfaces, and people touch the food, and it is open to contamination with dirt and microorganisms that cause food poisoning and food spoilage. So it is very important to keep these areas clean and to cover the food.

Utensils

Staff should thoroughly wash and disinfect utensils to keep them hygienic and stop them from spreading bacteria between foods. After cleaning, utensils should be

sanitized by heating to 77°C (171°F) for at least 30 seconds or by using chemical germicides at room temperature for at least 10 minutes (see Chap. 6 for more information about sanitizing). Cracked, chipped, or dented dishes or utensils can carry bacteria—and should be thrown away. Food and microorganisms can collect in cracks and dents, and cleaning and sanitizing may not remove them.

Servers should not touch any surface that will touch customers' mouths or the food. Microorganisms can transfer from hands and surfaces to dishes, utensils, or food, and then to consumers.

Employees' Health

Foodservice operations should not hire employees who have or are carriers of any disease that could be carried by food, water, or utensils. If workers have colds, flu, or skin infections, supervisors should give them responsibilities where they do not come in contact with food, or supervisors should not let them work until they are healthy.

Many local health codes require health checks every 6 months or 1 year. Employees may need a health card stating that they do not have an infectious disease. These examinations do not protect consumers against the short-term diseases that most workers experience every year, so it is important to train employees to report infections and visit their doctor when necessary. Training should also include personal hygiene to reduce the spread of disease (see Chap. 4).

Buffet or Cafeteria Service

The following can reduce contamination during buffet or cafeteria service:

- Cooling tables should keep foods at 2°C (36°F) or cooler so that microorganisms cannot grow quickly.
- Warming tables should keep foods at 60°C (140°F) or hotter.
- Authorized workers should serve food in hygienic conditions; customers should not handle serving utensils.
- Servers should wear hairnets or hats, and should not touch their hair while they are serving so that they do not contaminate the food.
- Operators should place transparent shields between the customers and the service tables and dishes so that customers cannot handle foods, or breathe, cough, or sneeze on foods.

Inspection Programs

The foodservice establishment should encourage local public-health authorities to inspect the facilities. If the local health department uses checksheets, the foodservice establishment should use a copy of these sheets for its own sanitation program. The operator should keep records of inspection reports and how the facility corrected any problems.

The Food and Drug Administration is promoting the Hazard Analysis Critical Control Points (HACCP) concept to stress factors that are known to cause foodborne disease. The National Restaurant Association (NRA) introduced a similar self-inspection tool, the Sanitary Assessment of Food Environment (SAFE), to the foodservice industry. Both tools have the same goal of consumer protection and stress safe food handling to reduce contamination and prevent pathogens from growing in foods.

County health departments throughout the United States are using the SAFE concept to inspect restaurants. SAFE looks at control of time and temperature at each step during preparation of foods that could be hazardous. (Foods should be kept hot or cold, and the duration of time the food is between 4°C (40°F) and 60°C (140°F) should be as short as possible (see Fig. 12.1.) SAFE protects foodservice establishments from outbreaks of foodborne disease by identifying, controlling, and monitoring critical points in the food-handling and preparation operation. This inspection program spends the most effort on foods that have the highest risk and on the production steps where the food is most likely to be contaminated. When operators have identified the critical control points, they can plan production and sanitation to make sure the food is safe.

SAFE is a simpler version of HACCP, and is based on three principles:

1. Keep it clean.
2. Keep it cold.
3. Keep it hot.

SAFE makes the foodservice establishment less likely to be held liable for outbreaks of foodborne illness because it can show that it has done all it could to keep the food safe. The first step in designing a SAFE program is to diagram each preparation step and record food temperatures every 30 minutes. A SAFE program should concentrate on moist, nonacid, high-protein foods, such as eggs, milk and dairy products, most meats, poultry, seafood, sauces and gravies, cooked cereal grains, and cooked vegetables.

A SAFE program should look for processing steps that tend to increase the risk of foodborne disease:

Multiple preparation steps: More handling means more chances to contaminate the food.

Temperature changes: Foods are at temperatures in the danger zone (4–60°C, 40–140°F) during heating and cooling.

Large volume: Large volumes of food take longer to heat and cool, so microorganisms have more time to grow.

Contaminated basic foods: Field dirt or pesticides can contaminate raw produce; raw red meats and poultry can be contaminated during slaughter; and raw seafood can carry viruses, bacteria, or parasites.

The SAFE program checks food items from the time they arrive at the receiving area until workers serve them to customers. Workers record temperatures and times at the beginning and end of each handling step. A SAFE survey identifies several control points, but only a few are critical control points. These include:

- Establishing good personal hygiene procedures and keeping equipment, utensils, and surfaces clean

- Holding foods at 2°C (36°F) or lower or 60°C (140°F) or higher
- Destroying microbes by cooking foods to an internal temperature hotter than 74°C (165°F)

Critical controls and monitoring points should be realistic. If a step does not meet its critical control goal (if the temperature is not cold enough or hot enough, if the temperature is in the danger zone for too long, or if cleaning and sanitation are not done properly), management may need to redesign the process.

CLEANING AND SANITIZING

The foodservice operator can choose from many cleaning and sanitizing methods and products. The operator needs to decide which methods and products will work best and must be certain that they are used properly.

Cleaning Basics

Foodservice establishments need a planned cleaning and sanitizing schedule and proper supervision. If workers are hurrying to meet the needs of customers, they often neglect parts of the sanitation program. The manager must understand food safety, be alert, and insist on disciplined sanitation. He or she must know how to be certain that conditions are sanitary. Workers must clean and sanitize all surfaces that touch food every time they are used, when there is a break in service, or at set times if employees use the surfaces all the time.

Cleaning uses the principles of chemistry in a practical way. Sanitarians should choose the best cleaning compound for each job. A compound that works well for one cleaning job may not be satisfactory for another cleaning job. See Chapter 5 for more information about cleaning compounds. Cheap cleaning compounds are not always more economical than those that are more expensive, because workers may need to use more of the cheaper compounds if they do not work as well as those that are expensive.

Alkaline cleaners do not work on some types of soil—for example, lime crusts on dishwashing machines, rust stains in washrooms, and tarnish on copper and brass. Acid cleaners, usually with a detergent, work well to remove these soils.

If soil is fixed so firmly to a surface that alkaline or acidic cleaners cannot remove it, a cleaner containing a scouring agent (usually finely ground feldspar or silica) usually works. Abrasives work well on worn and pitted porcelain, rusty metals, or very dirty floors. Workers should be careful with use of abrasives in a foodservice facility because they can scratch smooth surfaces.

Sanitizing Basics

Operators use heat or chemicals to sanitize equipment, surfaces, and utensils. Neither type of sanitizing works if staff do not first clean and rinse equipment, surfaces, and utensils properly. Any soil that stays on the surface can protect microorganisms from the sanitizer. (See Chap. 6 for more information about sanitizing.)

Because cooking utensils are heated during cooking, they may not seem to need to be sanitized. But heat from cooking does not always make every part of the utensil hot enough for long enough to make sure it is sanitized. During heat sanitizing, the temperature must be hot enough for long enough to kill microorganisms.

During chemical sanitizing, the clean object is immersed in the sanitizer for about 1 minute, or the surface is rinsed or sprayed with sanitizer at twice the normal strength. The strength of the sanitizing solution should be tested frequently, because the sanitizer is used up as it kills bacteria. The sanitizer should be changed when it is not strong enough to work properly. Firms that make sanitizers normally provide free test kits to check the strength of the sanitizer. Workers should only use toxic sanitizing agents on surfaces that do not touch the food.

Detergent-sanitizers contain a sanitizer and a cleaning compound. These products can sanitize, but sanitizing should be done in a separate step from cleaning because the sanitizing power can be lost during cleaning if the chemical sanitizer reacts with the soil. Detergent-sanitizers tend to be more expensive than plain cleaning compounds and have limited uses.

Employees can clean and sanitize most portable items that touch food at a washing area away from the food preparation area. The work station should have three or more sinks, separate drain boards for clean and soiled items, and an area for scraping and rinsing food waste into a garbage container or disposal. If the sanitizing step uses hot water, the third compartment of the sink must have a heating unit and a thermometer to make sure the water stays at about 77°C (171°F). Different areas in the facility may need to meet different regulations for cleaning and sanitizing.

Cleaning Steps

Manual cleaning and sanitizing of a typical foodservice facility has eight steps:

1. Clean sinks and work surfaces before each use.
2. Scrape off heavy soil and presoak to remove large amounts of soil that could use up the cleaning compound. Sort items and presoak silverware and other utensils in a special soaking solution.
3. Wash items in the first sink in a clean detergent solution at about 50°C (122°F), using a brush or dish mop to remove the soil.
4. Rinse items in the second sink. This sink should contain clear, clean water at about 50°C (122°F). This removes traces of soil and cleaning compounds that may stop the sanitizing agent from working properly.
5. Sanitize utensils in the third sink by dipping them in hot water (82°C, 180°F) for 30 seconds or in a chemical sanitizing solution at 40 to 50°C (104–112°F) for 1 minute. The sanitizing solution should be mixed at twice the recommended strength so that it does not get too dilute when items carry water from the rinse sink. It is important to get rid of air bubbles that could stop the sanitizer from getting inside items.
6. Air-dry sanitized utensils and equipment. Wiping can recontaminate them.

7. Store clean utensils and equipment in a clean area more than 20 cm (8 in) off the floor to protect them from splashing, dust, and contact with food.
8. Cover fixed equipment when not using it, especially the parts that touch the food.

Stationary Equipment

Food preparation equipment that cannot be moved should be cleaned according to the manufacturer's instructions for taking it apart and cleaning it. Here are some general steps:

1. Unplug all electrical equipment.
2. Take all equipment apart; wash and sanitize every part.
3. Wash and rinse all surfaces that touch food with a sanitizing solution at twice the strength used to sanitize by dipping.
4. Wipe all surfaces that do not touch food. Dip cloths used to wipe down equipment and surfaces in a sanitizing solution. Keep these cloths separate from other wiping cloths.
5. Air-dry all clean parts before putting them back together.
6. Follow the manufacturer's instructions for equipment that is designed to have detergent and sanitizing solutions pumped through it. Use high-pressure, low-volume cleaning equipment (see Chap. 7) and spray devices for sanitizing. To sanitize, spray with double-strength solution of sanitizer for 2 to 3 minutes.
7. Scrub wooden cutting boards with a nontoxic detergent solution and stiff nylon brush (or use a high-pressure, low-volume cleaning wand). Apply a sanitizing solution after every use. Replace worn, cut, or scarred wooden boards with polyethylene boards. Do not dip wooden cutting boards in a sanitizing solution.

Floor Drains

Workers must clean floor drains every day after cleaning everything else. Sanitation workers should wear heavy-duty rubber gloves to remove the drain cover and should use a drain brush to remove the debris. When they replace the cover, they should flush it with a hose going through the drain so that it does not splatter. Workers should pour a heavy-duty alkaline cleaner at the strength the manufacturer recommends down the drain, wash the drain with a hose or drain brush, and rinse it. Workers should use a quaternary ammonium plug (Quat Plug), or pour a chlorine or quat sanitizing solution down the drain.

Cleaning Tools

Foodservice operators should store cleaning tools separately from sanitizing tools. They should rinse, sanitize, and air-dry cloths, scrubbing pads, brushes, mops, and sponges after use. Clothing should be laundered every day. All buckets and mop pails should be emptied, washed, rinsed, and sanitized at least once a day.

Mechanized Cleaning and Sanitizing

Mechanized cleaning can work better than hand cleaning if it is used properly. More operations are beginning to understand the importance of good sanitation, and as operators prepare greater volumes of food, more of them are using dishwashing machines. Larger foodservice establishments may also use portable high-pressure, low-volume cleaning equipment.

There are two basic types of dishwashing machine: high-temperature washers and chemical sanitizing machines.

High-temperature washers. The four main types of high-temperature washer are discussed below. The sanitizing temperature for these washers should be at least 82°C (180°F), but not higher than 90°C (194°F).

1. *Single-tank, stationary-rack-type with doors.* This washer contains racks that do not move during the wash cycle. Utensils are washed using a cleaning compound and water at 62 to 65°C (144–149°F) sprayed from underneath the rack, but the machine may also have sprayers above the rack. After the wash cycle, the machine has a hot-water final rinse.
2. *Conveyor washer.* This equipment has a moving conveyor that takes utensils through the washing (70–72°C, 158–162°F), rinsing, and sanitizing (82–90°C, 180–194°F) cycles. Conveyor washers may contain a single tank (each step in the cycle is done one at a time) or multiple tanks (one each for washing, rinsing, and sanitizing).
3. *Flight-type washer.* This washer is a high-capacity, multiple-tank unit with a peg-type conveyor. It may have a built-on dryer and is common in large foodservice facilities.
4. *Carousel or circular conveyor washer.* This multiple-tank washer moves a rack of dishes on a peg-type conveyor or in racks. Some models stop after the final rinse.

Chemical sanitizing washers. The three main types of chemical sanitizing dishwasher are discussed below. Glassware washers are another type of chemical sanitizing machine. The water temperature during chemical sanitizing should be 49 to 55°C (120–131°F).

1. *Batch-type dump.* This washer washes and rinses in one tank. The machine times each step and releases the cleaning compound and sanitizer automatically.
2. *Recirculatory door-type, nondump washer.* The water does not completely drain from this washer between cycles. The machine dilutes the water with freshwater and reuses it during the next cycle.
3. *Conveyor-type, with or without a power prerinse.* This machine uses a conveyor to carry utensils and dishes from one cycle to the next and may or may not include a power prerinse.

When buying or using dishwashing equipment, it is important to consider the following:

1. Be certain that the dishwasher is the right size to handle the load.
2. If the equipment uses hot water to sanitize, it will need a booster heater that can provide water at 82°C (180°F).
3. Thermometers must be accurate to make sure that the water is at the right temperature.
4. It is important to install, maintain, and operate the equipment properly so that it cleans and sanitizes properly.
5. The layout of the dishwashing area should be convenient and efficient for staff and for the operation.
6. Operators should consider whether they need a prewash cycle instead of scraping and soaking very dirty utensils.
7. In machines with compartments, rinse-water tanks should be protected so that washwater cannot contaminate rinse water.
8. Staff should clean large dishwashers at least once a day following the manufacturer's instructions.

Table 17.1 gives some ideas for troubleshooting problems with dishwashers.

Cleaning-in-place (CIP) equipment. Equipment such as automatic ice-making machines and soft-serve ice-cream and frozen yogurt dispensers are designed so that staff can clean them by running a detergent solution, hot-water rinse, and sanitizing solution through the unit. Operators should look for machines that keep the cleaning and chemical sanitizing solution in a fixed system of tubes and pipes for a set amount of time. The cleaning water and solution should not be able to leak into the rest of the machine. The cleaning and sanitizing solutions must touch all surfaces that touch the food; the CIP equipment must drain itself; and operators must be able to look inside to make sure that the unit is clean.

Recommendations for Cleaning Specific Areas and Equipment

Area. Floors

Frequency. Daily

Supplies and equipment. Broom, dustpan, cleaning compound, water, mop, bucket, and powered scrubber (optional)

Daily

1. Clean all table surfaces (wipe pieces of food into a container, wash table with warm soapy water, rinse surfaces with clean water).
2. Stack chairs on tables or take them out of the area.
3. Sweep and remove trash using a push broom.
4. Put up signs to warn people that the floor is wet.
5. Mop floors or use a mechanical scrubber in larger operations. Use detergent (15 g/L or 2 oz/gal) and rinse with clear water.

TABLE 17.1. Dishwashing Problems and Solutions

Problem	Possible Cause	Suggested Solution
Dirty dishes	Not enough detergent	Use enough detergent in washwater to remove and suspend soil.
	Washwater too cool	Keep water at recommended temperature to dissolve food, and preheat for sanitizing step.
	Wash and rinse times too short	Allow enough time for wash and rinse steps to work (use an automatic timer or adjust conveyer speed).
	Bad cleaning	Unclog rinse and wash nozzles to keep the right pressure, spray pattern, and flow. Make sure overflow is open. Prescrape dishes to keep washwater as clean as possible. Change water in tanks more often.
	Bad racking	Make sure that racks are loaded properly, based on the size and type of utensils. Presoak silverware and place in silver holders without sorting or shielding.
Films	Hard water	Use a water softener outside the machine. Use a detergent that contains a water conditioner. Make sure that washwater and rinse water are not too hot (hot water may cause film to settle out).
	Detergent film	Make sure that there is enough rinse water at high-enough pressure. Make sure that wash jets are not worn or spraying at the wrong angle and getting into rinse water.
	Poor cleaning or rinsing	Clean the machine often and carefully so that scale does not build up inside it. Make sure that there is enough water with enough pressure.
Greasy films	Too acidic (low pH); not enough detergent; water too cool; equipment not cleaned properly	Make sure that cleaning solution is alkaline enough to suspend grease. Check cleaning compounds and water temperatures. Unclog wash and rinse nozzles so they can spray properly. Change water in tanks more often.
Foam	Detergent type or solids suspended in water	Use a low-sudsing detergent. Reduce the amount of solids in the water.

Table 17.1 *(Continued)*

Problem	Possible Cause	Suggested Solution
Foam *(continued)*	Very dirty dishes	Scrape and prewash dishes so that food does not decompose and cause foam.
	Equipment not cleaned properly	Make sure that spray and rinse nozzles are open. Keep equipment free of deposits and films that can cause foam.
Streaks	Alkaline water, solids dissolved in water	Treat the water to make it less alkaline. Use a rinse additive. Treat the water outside the machine if the problem is severe.
	Bad cleaning or rinsing	Make sure that there is enough rinse water at high-enough pressure. Rinse alkaline cleaners off dishes thoroughly.
Spots	Hard rinse water	Soften the water inside or outside the machine. Use rinse additives.
	Rinse water too hot or too cold	Check rinse-water temperature. Dishes may be flash drying, or water may be drying on dishes rather than draining off.
	Not enough time between rinsing and storage	Change to a low-sudsing cleaner. Treat the water to reduce the solid content.
	Food soil	Remove most of the soil before washing. Change water in the tanks more often.
Stains: coffee, tea,or metal	Wrong detergent	Use a detergent with chlorine.
	Equipment not cleaned properly	Make sure spray and rinse nozzles are open. Keep equipment free of deposits and films.

Source: National Sanitation Foundation (1982).

Weekly

1. Do the daily cleaning.
2. Scrub floors (use a powered scrubber and/or buffer on floor, rinse with clean water at 40–55°C [104–131°F] water. Sponge-mop and dry-mop the floor).

Area. Walls

Frequency. Daily and weekly

Supplies and equipment. Handbrush, sponge, cleaning compound, bucket, water, and scouring powder

Daily

1. Clean spots that look dirty. Use cleaning compound (15 g/L or 2 oz/gal), hand-wipe dirty areas, rinse with clean water, wipe dry.

Weekly

1. Remove debris from walls.
2. Mix 15 g of cleaning compound per liter of water (2 oz/gal).
3. Scrape walls using a handbrush. Scrub tiles and grout.
4. Rinse wall surfaces with clean, warm water.
5. Wipe dry using clean cloths or paper towels.

Area. Shelves

Frequency. Weekly

Supplies and equipment. Handbrush, detergent, sponge, water, and bucket

1. Remove items from the shelves, and store on a pallet or other shelves.
2. Brush off all debris into a pan or container.
3. Clean shelves in sections (scrub with detergent and warm water).
4. Replace items on the shelves (throw away damaged goods).
5. Mop floor area under shelves with a clean, damp mop.

Equipment. Stack oven

Frequency. Clean once a week thoroughly; wipe daily

Supplies and equipment. Salt, metal scraper with long handle, metal sponges, cleaning compound in warm water, 4-liter (1-gal) bucket, sponges, stainless-steel polish, ammonia, vinegar, or oven cleaner if appropriate

Weekly

1. Turn off the heat and scrape the inside. Sprinkle salt on hardened oven spills. Turn the oven on at 260°C (500°F). When the spills have completely carbonized (blackened), turn off the oven. Cool thoroughly. Scrape the floor with a long-handled metal scraper. Use a metal sponge or hand scraper on the inside of doors, handles, and edges.
2. Brush out scraped carbon and other debris. Begin with top deck of stack oven. Brush out with stiff bristle brush, and use dustpan to collect debris.
3. Wash doors (use a hot detergent solution only on enamel surfaces, rinse, and wipe dry).
4. Brush inside using a small broom or brush.
5. Clean and polish outside of oven (wash the top, back, hinges, and feet with warm cleaning compound solution, rinse, and wipe dry; polish all stainless steel).

Note: Never pour water on the outside of the oven, and do not use a wet cloth or sponge on the outside surfaces. Do not sponge, drip, or pour water inside the oven to clean it.

Equipment. Hoods

Frequency. Weekly (at least)

Supplies and equipment. Rags, warm soapy water, stainless-steel polish, degreaser for filters

1. Remove filter, carry it outside, rinse with a degreaser, and run it through the dishwasher after all dishes and eating utensils have been cleaned.
2. Wash inside and outside of hood (use warm, soapy water and a rag to wash hoods completely on the inside and outside to remove grease; clean drip trough in area below filters).
3. Shine hood with polish (spray polish on hood and wipe off; use a clean rag on inside and outside).
4. Replace filters.

Equipment. Range surface unit

Frequency. Weekly

Supplies and equipment. Putty knife, wire brush, damp cloth, hot detergent solution, 4-liter (1-gal) container, vinegar or ammonia as appropriate

1. To clean back apron and warming oven or shelf, let surfaces cool. Use a hot, damp cloth wrung almost dry. Remove hardened dirt with a putty knife, and scrape edge of plates. Scrape burned material from top surfaces with a wire brush.
2. Lift plates. Remove burned particles with a putty knife, and scrape edge of plates. Scrape burned material from flat surfaces with a wire brush.
3. Wipe heating elements with a damp cloth.
4. Clean base and exterior (wipe with a cloth dampened with hot detergent water).
5. Clean grease receptacles and drip pans (soak grease receptacles and drip pans in a detergent solution for 20 to 30 min, scrub, rinse, and dry).

Note: Do not dip heating elements in water.

Equipment. Griddles

Frequency. Daily

Supplies and equipment. Spatula, pumice stone, paper towels, hot detergent solution

1. Turn off heat. Remove grease after each use. When surface is cool, scrape it with a spatula or pancake turner. Wipe clean with dry paper towels. Use a pumice stone block to clean burned areas on plates. (Do not use pumice stone daily.)
2. Clean grease and drain troughs (pour a hot detergent solution into a small drain and brush; rinse with hot water).

3. Empty grease receptacles, and remove grease with hot detergent solution; rinse and dry.
4. Scrub guards, front, and sides of the griddle using a hot detergent solution. Wash off grease, splatter, and film. Rinse and dry.

Equipment. Rotary toaster

Frequency. Daily

Supplies and equipment. Warm detergent solution, brush, cloths, stainless-steel polish, nonabrasive cleaner

1. After the toaster is cooled, disconnect it and take it apart (remove pan, slide, and baskets).
2. Clean surface and underneath using a soft brush to remove crumbs from the front surface and behind bread racks.
3. Clean frame and inside as far as you can reach. Wipe clean with a warm hand-detergent solution. Rinse and dry. Polish if necessary with a nonabrasive cleaning powder. The outside casing should not collect grease or dirt. Do not let water and cleaning compounds touch the conveyor chains. Polish the frame if it is made of stainless steel.

Equipment. Coffee urns

Frequency. Daily

Supplies and equipment. Outside cleaning compound (stainless-steel polish), inside cleaning compound (baking soda), urn brush, faucet, and glass brush

1. Rinse urns by flushing with cold water.
2. Be certain that the outer jacket is three-quarters full of water. Turn on heat. Open water inlet valve and fill coffee tank with hot water to the coffee line. Add recommended quantity of cleaning compound (10–15 g/L, 1.3–2.0 oz/gal). Allow solution to remain in the liner for approximately 30 min with the heat on full.
3. Brush the liners, faucet, gauge glass, and draw-off pipe (scrub inside of tank, top rim, and lid); draw off 2 L (1/2 gal) of solution, and pour it back to fill the valve and sight gauge. Put the brush in the gauge glass and coffee draw-off pipe, and brush briskly.
4. Drain by opening the coffee faucet and completely draining the solution. Close the faucet.
5. To rinse the machine, open the water inlet valve into the coffee tank. Use 4 L (1 gal) of hot water. Open the faucet for 1 min to allow water to flow and sterilize the dispenser
6. Take faucet apart and thoroughly scrub with a brush. Rinse spigot thoroughly.
7. Twice weekly, make a solution of 1 cup baking soda in 4 L (1 gal) hot water, and hold in the urn for approximately 15 min. Drain. Flush thoroughly with hot water before use.

Note: Put a tag on the faucet while the urn is soaking with the cleaning compound.

Biweekly

1. Fill urn with a destaining compound solution (2 tablespoons to 20 L [5 gal] of water at 80°C [176°F] or as directed by manufacturer).
2. Open spigot and draw off 4 L (1 gal) of solution. Pour this back into the tank, and let it stand for 1 hr at 75 to 80°C (167–176°C).
3. Scrub liner and gauge glass using a long-handled brush to loosen scale.
4. Clean faucet (take faucet valve apart and clean each piece; soak in hot water).
5. Rinse valve parts and inside of urn 3 or 4 times with hot water until all traces of cleaning compound are removed; put back together.
6. Refill urn.

Note: To remove stain from vacuum-type coffeemakers, use a solution of 1 teaspoon of compound per liter (quart) of warm water. Fill the lower bowl up to within 5 cm (2 in) of the top, and put the unit together.

Equipment. Iced-tea dispensers

Frequency. Daily

Supplies and equipment. Rags and warm, soapy water

1. Clean outside with a damp cloth.
2. Empty drip pan and wash drip pan and grill with a mild detergent and warm water.
3. Wash trough (open front jacket, remove mix trough, and wash in a mild detergent and warm water).
4. Wash plastic parts (do not soak plastic parts in hot water or wash in dishwashing machines).

Equipment. Steam Tables

Frequency. Daily

Supplies and equipment. Dishwashing detergent, spatula, scrub brush, and cloths

1. Turn off heating unit by turning steam valve counterclockwise for steam-heated food or turning dial to OFF position for electrically heated food.
2. Remove insert pans and take them to the dishwashing area after each use. Hand-clean, sanitize, and air-dry. Store in a clean area until needed.
3. Drain water from the steam table. Remove the overflow pipe, using a cloth to prevent injury.
4. Prepare the cleaning solution (30 ml [1 fl oz] of dishwashing liquid in a suitable container).

5. Scrape out food particles from the steam table with a spatula or dough scraper.
6. Scrub inside, and clean outside with a scrub brush and cleaning solution.
7. Rinse outside using enough clear water to remove all of the detergent.

Note: Mobile electric hot-food tables: Clean corrosion-resistant steel after each use. Remove ordinary deposits of grease and dirt with mild detergent and water. Rinse thoroughly and dry.

Equipment. Refrigerated salad bars (with ice beds or electrically refrigerated)

Frequency. After each use

Supplies and equipment. Detergent, plastic brush, and sanitizing agent

1. Transfer shallow pans or trays and containers to preparation areas after each meal. Run insert pans and/or trays through dishwashing machine.
2. Clean table counter using detergent and plastic brush. Rinse and sanitize by swabbing with a solution containing a sanitizing agent.
3. Descale whenever needed to prevent rust, lime, or hard-water deposits in those with ice beds. Fill table bed with boiling water, and add a descaling compound following the manufacturer's recommendations. Allow to stand for several hours. Scrub with a plastic brush. Drain. Rinse thoroughly. Sanitize by spraying with a sanitizing solution.
4. Defrost electrically refrigerated units as often as needed. Follow up with a cleaning procedure as described above.

Equipment. Milk dispenser

Frequency. Daily

Supplies and equipment. Sanitized cloth or sponge, mild detergent, sanitizing agent

1. To remove empty cans from the dispenser, place a container under the valve, open the valve, and tip the can forward in the dispenser to drain out the remaining milk. Remove the tube and lift out the oar.
2. Wipe up any spills using a sanitized cloth or sponge.
3. Wash the whole inside surface with a milk cleaning solution and rinse.
4. Clean the outside using a stainless-steel cleaner. If the steel is discolored or stained, swab with a standard chemical, leave for 15 to 20 min, rinse with clear water, and polish with a soft cloth.
5. Take apart and clean the valves every day or each time empty cans are removed. Wash in detergent water. Rinse and sanitize.
6. Wipe the bottom of full milk cans with a sanitizing solution before placing in the dispenser.

Equipment. Deep fat fryer

Frequency. Daily

Supplies and equipment. Knife, spatula, wire brush, detergent, long-handled brush, vinegar, nylon brush, dishwashing compound

Daily

1. Turn off the heating element. Allow fat to cool to 65°C (149°F).
2. Drain and filter the fat after each use. Open drain valve, drain entire kettle contents, and filter into a container. Remove fat container. Place a clean fat container into the well or wash, and replace the original container.
3. Remove baskets, and scrape off the oxidized fat with a knife. Remove loose food particles from the heating units with a spatula or a wire brush. Flush down sides of the kettle with a scoop of hot fat. Soak basket and cover in a deep sink with hot detergent.
4. Close the drain. Fill the tank with water up to the fat level; add 60 ml (2 fl oz) of dishwashing compound.
5. Turn on heat to 121°C (250°F) and boil 10 to 20 min.
6. Turn off heat, open drain, and draw off cleaning solution.
7. Scrub inside using a long-handled brush. Flush out with water. Clean the basket with a nylon brush, and place it back in the kettle.
8. Fill the kettle with water to rinse. Add 1/2 cup of vinegar to neutralize the remaining detergent. Turn on the heating element. Boil 5 min to sanitize and *turn off heat.* Drain. Rinse with clear water.
9. Air dry all parts, including baskets and strainer.
10. When kettle is cool, wipe off the outside with a grease solvent or a detergent solution. Rinse.

Weekly

1. Fill kettle with water to fat level. Heat to at least 80°C (176°F) or boil for 5 to 10 min. Turn off the heat.
2. Add 1/2 tablespoon of destaining compound (stain remover for tableware) per liter (quart) of water. Stir solution, and loosen particles on sides of kettle.
3. Place screens and strainers in water at 80°C (176°F) containing 1/2 tablespoon of destaining compound per liter (quart). Allow to stand overnight. Rinse thoroughly and air-dry.
4. Drain kettle and rinse thoroughly before replacing cleaned screen and strainer.

Equipment. Vegetable chopper

Frequency. Daily

Supplies and equipment. Brush, sponge, cloth, bucket, detergent, sanitizer solution

1. Take parts apart after each use (turn off power first, and wait until knives have stopped revolving).

2. Clean knives, bowl guard, and bowl. Remove blades from the motor shaft and clean carefully. Wash with a hand detergent solution. Rinse and air-dry. Remove all food particles from the bowl guard. If the bowl is removable, wash it with other parts; if the bowl is fixed, wipe out food particles from table or base. Clean with a hand detergent solution; rinse and air-dry.
3. Clean parts and under chopper surface. Dip small parts in a hot hand detergent solution; wash, rinse, and air-dry.
4. Put the parts back together. Replace comb in guard. Attach bowl to the base and knife blades to the shaft. Drop guard into position.

   Note: The mechanical details of choppers vary a lot.

Equipment. Meat slicer

Frequency. Daily

Supplies and equipment. Bucket, sponge, cloth, brush, detergent, sanitizer solution

1. Disconnect the equipment. Remove meat holder and chute by loosening screw. Remove scrap tray by pulling it away from the knife. Remove the knife guard. Loosen bolt at the top of knife guard in front of the sharpener. Remove bolt at the bottom of the knife guard behind chute. Remove guard.
2. Clean slicer parts by scrubbing in a sink filled with hot detergent solution. Rinse with hot water. Immerse in a sanitizer solution. Air-dry.
3. Wipe the knife blade with a hot detergent solution. Wipe from center to edge. Air-dry.
4. You may use a sanitizer block to sanitize the blade. This block is sliced by the blade after the slicer is put back together.
5. Clean receiving tray and underneath tray with a hot detergent solution. Rinse in hot clear water. Air-dry.

   Note: Do not pour water on equipment or immerse this equipment in water.

SANITATION MANAGEMENT AND TRAINING

Proper cleaning does not happen without good management. For a good cleaning program, management must make an effort to understand the type of cleaning the food-service establishment needs and to provide the right equipment and chemicals for the jobs. A good cleaning program makes sure that cleaning jobs are not forgotten, forces the sanitation manager to plan ahead and make the best use of resources, helps new employees with cleaning routines, gives a basis for inspections, and saves employees from having to decide what to clean when. Table 17.2 shows part of a cleaning schedule for a food preparation area. A full cleaning schedule could use the same format. Management should arrange the schedule logically so that nothing is forgotten.

Management should schedule most cleanups when they are least likely to contaminate foods or interfere with service. Staff should not vacuum and mop while food is being prepared and served, but they should clean as soon as possible afterwards so that soil does not dry and harden and bacteria do not have a chance to grow.

TABLE 17.2. Part of a Sample Cleaning Schedule for Food Preparation Area

Item	When	What	Use	Who
Floors	As soon as possible	Wipe up spills	Cloth, mop and bucket, broom and dustpan	_____
	Once per shift between rushes	Damp mop	Mop, bucket, or mechanical scrubber	_____
	Weekly, Thursday evening	Scrub	Brushes, bucket, detergent	_____
	January, June	Strip, reseal	See procedure	_____
Walls and ceilings	As soon as possible	Wipe up splashes	Clean cloth; portable high-pressure, low-volume cleaner; or portable foam cleaner	_____
	February, August	Wash walls		Contracted specialists
Work tables	Between uses and at end of day	Clean and sanitize tops	See cleaning procedure for each table	_____
	Weekly, Saturday	Empty, clean, and sanitize drawers; clean frame, shelf	See cleaning procedure for each table	_____
Hoods and filters	When necessary	Empty grease traps	Container for grease	_____
	Daily, closing	Clean inside and out	See cleaning procedure	_____
	Weekly, Wednesday evening	Clean filters	Dishwashing machine	_____
Broiler	When necessary	Empty drip pan,wipe down	Container for grease; clean cloth	_____
	After each use	Clean grid tray, inside, outside, top	See cleaning procedure for each broiler	_____

Adapted from *Applied Foodservice Sanitation,* 4th ed. Copyright 1992 by the Educational Foundation of the National Restaurant Association.

Managers should arrange the schedule so that cleaning times are evenly spaced and tasks are done in the right order.

Managers should discuss new cleaning programs with employees at a meeting and should demonstrate use of new equipment and procedures. It is important to explain why the program is being changed, how it will benefit the operation, and why employees should follow procedures exactly.

Managers should evaluate the sanitation program to make sure it is working properly. They may evaluate the program during on-going supervision and inspections. They must monitor the program to make sure that each step is followed. Managers should keep written records to show that employees are following the program.

Employee Training

Training takes workers and management away from the job. To make it worthwhile, many operations employ training specialists. Printed sheets, posters, demonstrations, slides, and films are all helpful in training.

It is difficult to prove that sanitation training is cost-effective. The operation cannot always measure the benefits. But sanitation could prevent a costly outbreak of foodborne illness or save an establishment from having to close until local health standards are met. It also improves the image of the operation, which tends to increase sales.

Employee training is important because it is difficult to hire capable and motivated workers. Training should be ongoing because the foodservice industry has a high rate of employee turnover, and employees need frequent reminders to keep good sanitation on their minds.

On-the-job training can work for some tasks but is not complete enough for sanitation training. Each employee involved in foodservice should understand what sanitation means and what he or she needs to do in his or her job to keep conditions sanitary.

The best way to train employees in a large or medium-sized firm is to set up a training department and hire a training director. Foodservice trainers have established their own professional association, the Council of Hotel and Restaurant Trainees.

In most small foodservice operations, supervisors are responsible for sanitation training. A trained employee or someone certified in foodservice sanitation should do the training.

Management can tell how well a training program works by whether employees can follow the sanitation program. The number of guest complaints and customer return rates also show if the training program works.

The National Restaurant Association Educational Foundation (1992) recommends two methods to evaluate training. An objective method uses tests or quizzes to find how much employees understand. In the other method, managers evaluate employees' job performance. Training works better when managers praise employees and use wall charts, pins, and certificates for good performance. Organizations such as the National Restaurant Association Educational Foundation and some regulatory agencies have certification courses that give training and recognition.

SUMMARY

- Today, food is handled more, so it is harder to protect it from contamination.
- Hygienic design of the facility and equipment makes cleaning easier in foodservice establishments.
- Several organizations and manufacturers have equipment standards that make it easier to choose sanitary equipment.
- The facility, equipment, and utensils must be thoroughly cleaned and sanitized.
- Mechanized cleaning using a dishwasher is a good way to clean and sanitize equipment and utensils, provided the dishwasher is well maintained and properly operated.
- Managers need a written cleaning and sanitizing program . They must train, supervise, and evaluate how well employees carry out the program.

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STUDY QUESTIONS

1. List three basic food safety procedures for foodservice operations.
2. What should a supervisor do if an employee has a cold, flu, or skin infection?
3. List three ways to make sure buffet or cafeteria service is safe.
4. What are four things that increase the risk that food will be contaminated during preparation?
5. Why must workers do cleaning and sanitizing in two separate steps, even if they use a combined detergent-sanitizer compound?
6. How should employees dry equipment, utensils, and surfaces after they wash and sanitize them and why?
7. Give two reasons why sanitation training needs to be ongoing.

TO FIND OUT MORE ABOUT SANITARY FOOD HANDLING IN FOODSERVICE

1. Contact equipment manufacturers or suppliers of cleaning compounds for information about how to clean and sanitize specific pieces of equipment, utensils, or surfaces.
2. The following organizations have books, booklets, posters, videos, and training manuals on sanitary food handling in foodservice:

- The Culinary Institute of America, Video Sales Office, 1433 Albany Post Road, Hyde Park, NY 12538-1499, (800) 285-8280.
- The Educational Foundation of the National Restaurant Association, 250 S. Wacker Drive, Suite 1400, Chicago, IL 60606-5834.
- Food and Drug Administration, 200 C Street SW, HFS-555, Washington, DC 20204, (202) 401-3532.
- Food Marketing Institute, 800 Connecticut Avenue, NW, Suite 500, Washington, DC 20006-2701, (202) 429-8298.
- Colorado State University, Department of Food Science and Human Nutrition, Fort Collins, CO 80523, (303) 491-7334.
- Cornell Cooperative Extension, Department of Food Science, 11 Stocking Hall, Ithaca, NY 14853, (607) 255-7922.
- North Carolina Cooperative Extension Service, Box 7624, Raleigh, NC 27695, (919) 515-2956.
- Ohio State University Extension Service, Department of Human Nutrition and Food Management, Ohio State University, 1787 Neil Avenue, 265 Campbell Hall, Columbus, OH 43210-1295, (614) 292-0827.
- Ohio University, Department of Environmental Health & Safety, Hudson Health Center 212, Athens, OH 45701-2991, (614) 593-0022.
- University of Massachusetts Cooperative Extension, 202 Chenoweth Laboratory, University of Massachusetts Cooperative Extension, Amherst, MA 01003, (413) 545-0552.
- Golden Gate Restaurant Association, 720 Market Street, Suite 200, San Francisco, CA 94103, (415) 781-3925.
- National Environmental Health Association, 720 S. Colorado Boulevard, Suite 970, South Tower, Denver, CO 80222, (303) 756-9090.
- National Live Stock and Meat Board, 444 North Michigan Avenue, Chicago, IL 60611, (800) 368-3138.
- Seattle King County Department of Health, Environmental Health Division, Smith Tower, 506 Second Avenue, Room 201, Seattle, WA 98104-2311, (206) 296-4722.
- Local state and county health departments. Check your telephone directory.



HOW TO AVOID COMMON FOOD SAFETY MISTAKES

In foodservice operations, the most common food safety mistakes are abuse of time and/or temperature, cross-contamination, and poor personal hygiene.

Time/temperature abuse

Foods need to be refrigerated (colder than 2°C, 36°F) or kept hot (hotter than 60°C, 140°F).

- Use digital thermometers with long probes to check the temperature inside the food. Some new thermometers even contain memory chips that keep records of temperatures at different times and on different days. Check oven and refrigerator thermometers.
- Check temperatures of food deliveries to make sure they have been kept cold on the truck. Don't let chilled foods sit out of the refrigerator after they are delivered.
- Cool hot foods in shallow pans using an ice bath or cooling paddle, or use ice as an ingredient instead of water before putting the foods in the refrigerator. Do not cool at room temperature. Putting hot foods in the refrigerator warms up other foods and may not cool the food fast enough.
- Cook foods to temperatures recommended in the FDA Food Code, and reheat foods to at least 74°C (165°F). Do not reheat foods more than once. After cooking or reheating foods, keep them hotter than 60°C (140°F).
- Prepare foods in small batches. Do not leave large amounts of ingredients sitting out at room temperature while you prepare them.

Cross-contamination

- Buy plenty of color-coded boards, and use each board for one specific type of food—e.g., chicken only, vegetables only, bread only. Wash boards in hot water, and sanitize after each use. If boards go black, they have bacteria growing on them—throw them out!
- Use nonabsorbent, washable mats to anchor cutting boards instead of towels that can absorb juices. Or replace towels between cutting jobs.
- Have specific knives for each type of food; clean and sanitize them between each cutting job. Label the drawers where the knives are kept so that the right knife is used for the right food.
- Wipe down the slicer blade with a clean, hot cloth between jobs, and sanitize. Use an antiseptic block—a block of solidified sanitizer that you slice on the slicer.
- Clean and sanitize the counter between cutting jobs.

- In the refrigerator, store on top shelves cooked foods and foods to be served raw, and uncooked raw foods on bottom shelves, so that contaminated juices do not drip onto ready-to-eat foods.
- On buffets, clean and sanitize or replace tongs, ladles, and spoons every half hour.

Poor Personal Hygiene

- Make sure all employees wash their hands thoroughly and often.
- Make sure employees have convenient handwashing stations so that they are not washing their hands in food preparation sinks (which may contain clean food) or in the washroom (which may have contaminated door handles).
- Train employees to scrub their hands for 30 seconds using antibacterial soap, a nail-brush, and hot water.
- Use plastic gloves properly. Employees must still wash their hands and must change gloves between touching different foods.

Source: Lorenzini, B. 1995. Avoid common food safety mistakes. *Restaurants & Institutions.* 105:122.